



SmartTrip Planner

AI-Powered Sustainable Travel Planning Platform

Week 4 Submission — Frontend-Backend Integration

Student: **Srishti** | Roll No: **26100787**

Course: **B.Tech CSE — 4th Year**

Date: **28 June 2026**

☒ **Full-Stack Application — React + FastAPI**

SmartTrip Planner is an AI-powered travel planning platform that helps users plan eco-friendly trips. It features a React frontend with Vite, a FastAPI backend with JWT authentication, and real-time data visualization with Recharts. The project follows a decoupled REST API architecture.



Frontend

- React 18** — UI framework with hooks
- Vite** — Build tool & dev server
- Tailwind CSS** — Utility-first styling
- Recharts** — Data visualization
- React Router** — SPA navigation
- Lucide React** — SVG icon library



Backend

- FastAPI** — Python REST API framework
- Uvicorn** — ASGI web server
- PyJWT** — JSON Web Token auth
- Passlib + Bcrypt** — Password hashing
- Pydantic** — Data validation
- In-memory store** — Data persistence

✓ JWT-secured authentication

✓ Real-time eco-score tracking

✓ Dark/Light theme support

✓ Search & filter trips

✓ Full CRUD trip management

✓ Interactive analytics charts

✓ Responsive mobile design

✓ AI travel recommendations

All API endpoints are secured with JWT Bearer token authentication. The frontend sends requests via a centralized API client (api.js) that automatically attaches the token from localStorage.

● API Endpoints — All returning 200 OK

Method	Endpoint	Status	Description
POST	/api/auth/login	200	Authenticate user, returns JWT token
GET	/api/auth/me	200	Get current authenticated user info
GET	/api/dashboard/stats	200	Aggregated trip statistics
GET	/api/dashboard/activity	200	Monthly activity & sustainability data
GET	/api/trips	200	List all trips for current user
POST	/api/trips	201	Create a new trip
GET	/api/trips/{id}	200	Get single trip by ID
GET	/api/trips/search?q=	200	Search trips by destination/tags
PUT	/api/trips/{id}	200	Update an existing trip
DELETE	/api/trips/{id}	204	Delete a trip

Users authenticate via email/password. On success, the backend returns a JWT token stored in localStorage. The AuthContext provider manages session state across the entire application.



Your Sustainable Travel Journey Starts Here

Join thousands of conscious travelers who plan smarter and explore responsibly with AI-powered itineraries.

- Personalized AI-generated itineraries
- Eco-friendly travel recommendations
- Save and manage your travel plans
- Track your sustainability impact

[Sign In](#)[Sign Up](#)

Sign in to access your travel dashboard



OR

Email Address

✉ you@example.com

Password

[Forgot password?](#)

🔒 ••••••••



☐ Remember me for 30 days

[Sign In](#) →

Demo: srishti@example.com / password123

By continuing, you agree to our [Terms of Service](#) and [Privacy Policy](#).

Dashboard — Overview Tab

Stats cards, charts, and trip list fetched from backend API

The dashboard loads data from 3 API endpoints simultaneously: `/api/dashboard/stats` (stat cards), `/api/dashboard/activity` (charts), and `/api/trips` (trip list). All data is real and computed from the backend database.

Welcome back, Srishti 🙌

Here's your travel overview for June 2026

[+ New Trip](#)[Overview](#)[My Trips](#)[Eco Impact](#)[Analytics](#)+12%**6**

Total Trips

+8%**6**

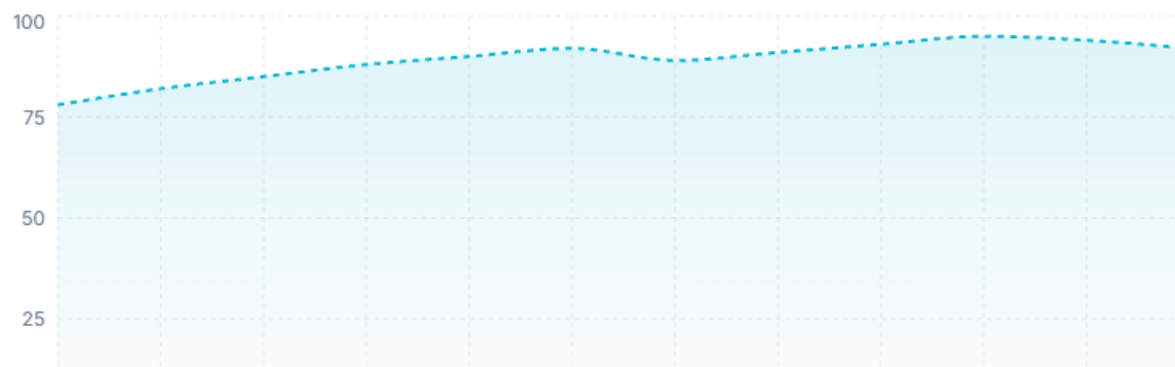
Destinations

+23%**14.0t**CO₂ Saved+5pt**94**

Eco Score

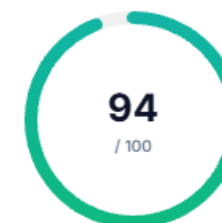
Trip Activity

Monthly trips & eco score trends

Trips Eco Score

Eco Impact

Sustainability breakdown

Excellent

Transport 35%

Accommodation 25%

The "My Trips" tab displays all user trips fetched from GET /api/trips. Each trip shows destination, duration, budget, eco-score, and tags. Users can delete trips (DELETE /api/trips/{id}) directly from here.

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↗ +12%

6

Total Trips


↗ +8%

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Destinations


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14.0t

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↗ +5pt

94

Eco Score

All Itineraries

Your travel plans



Bali, Indonesia

Beach

Culture

🕒 7 days • 💰 \$1,200 • 📅 Dec 2025

🌿 94



Swiss Alps, Switzerland

Adventure

Nature

🕒 5 days • 💰 \$2,100 • 📅 Nov 2025

🌿 88



Kyoto, Japan

Heritage

Culture

🕒 6 days • 💰 \$1,800 • 📅 Oct 2025

🌿 96



Upcoming

Next adventures



Tokyo, Japan

Jun 25 – Jul 1, 2026

CONFIRMED

🕒 7 days • 💰 \$2,400 • 🌿 91

Temple Visits

Street Food Tour

Mt. Fuji Day Trip



Patagonia, Chile

Jul 15 – Jul 24, 2026

PLANNING

🕒 10 days • 💰 \$3,200 • 🌿 95

The Eco Impact tab filters and displays only high eco-score trips (≥ 90). The sustainability breakdown shows environmental metrics across Transport, Accommodation, Activities, and Food categories.

Welcome back, Srishti 🙌

Here's your travel overview for June 2026

[+ New Trip](#)[Overview](#)[My Trips](#)[Eco Impact](#)[Analytics](#)↗ +12%**6**

Total Trips

↗ +8%**6**

Destinations

↗ +23%**14.0t**CO₂ Saved↗ +5pt**94**

Eco Score

Top Eco Trips

Trips with eco score ≥ 90

**Bali, Indonesia**

Beach

Culture

🕒 7 days • 💰 \$1,200 • 📅 Dec 2025🌿 94**Kyoto, Japan**

Heritage

Culture

🕒 6 days • 💰 \$1,800 • 📅 Oct 2025🌿 96**Costa Rica**

Eco-Tourism

Wildlife

🕒 8 days • 💰 \$1,500 • 📅 Sep 2025🌿 97

Upcoming

Next adventures

**Tokyo, Japan**

CONFIRMED

📅 Jun 25 – Jul 1, 2026🕒 7 days • 💰 \$2,400 • 🌿 91

Temple Visits

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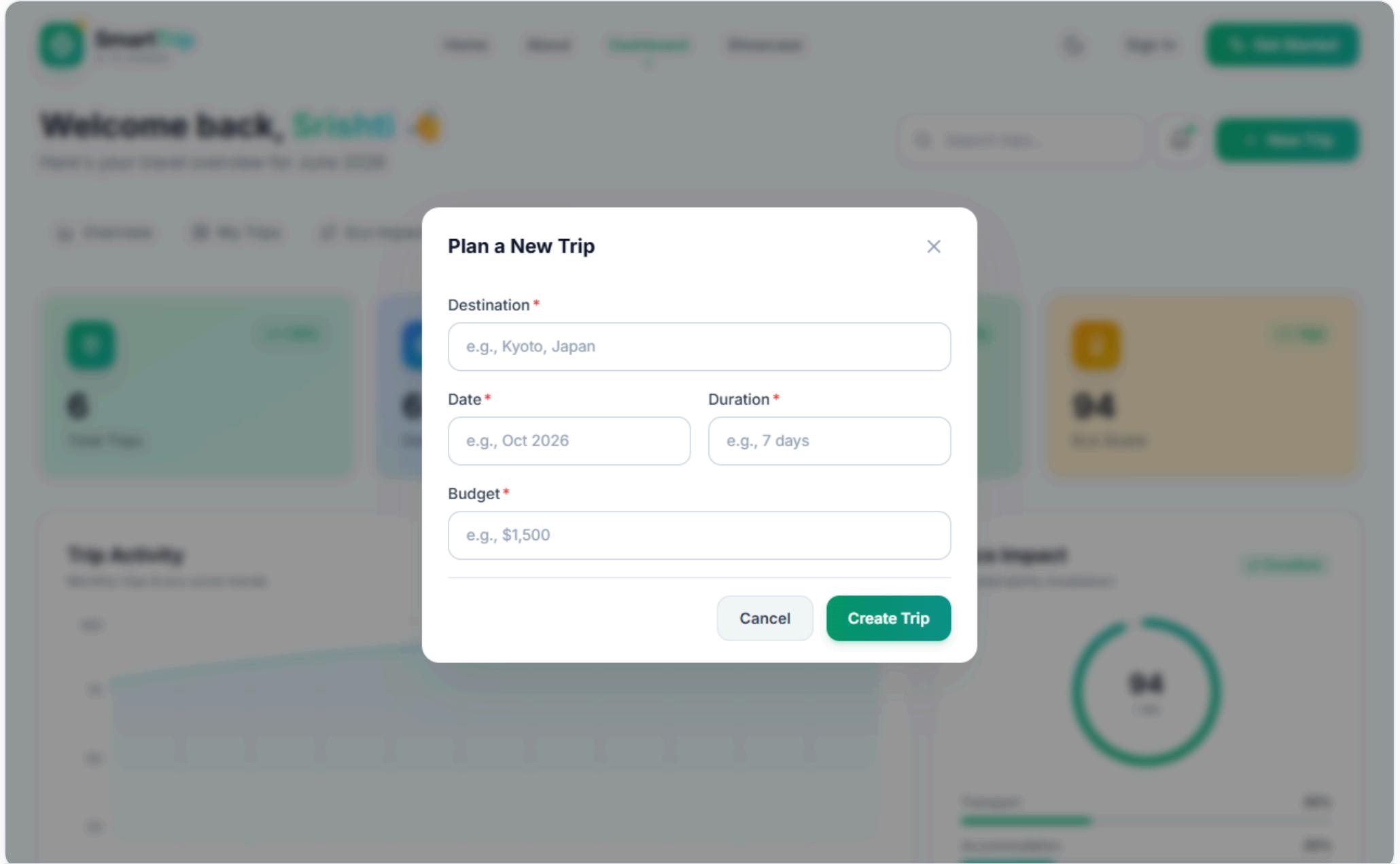
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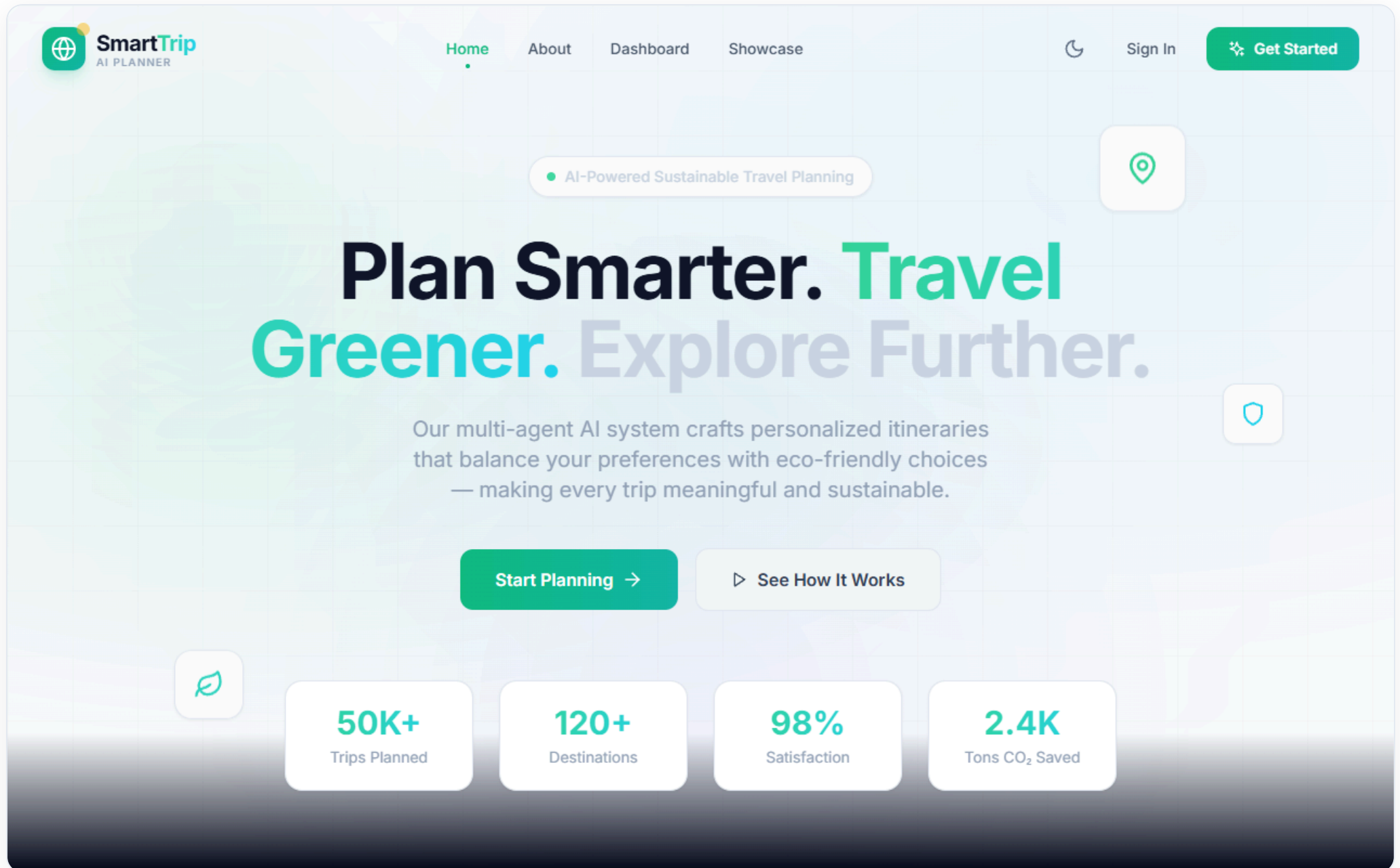
7

Create New Trip

Modal form with POST /api/trips integration

The "New Trip" modal collects destination, date, duration, and budget. On submission, it sends a POST request to /api/trips with the data. The backend validates the input via Pydantic models, creates the trip, and returns it. The UI immediately updates without a page refresh.





[About SmartTrip Planner](#)

Reimagining Travel with **AI** & Purpose

We're building the future of travel planning — where artificial intelligence meets sustainability to create meaningful journeys that benefit both travelers and the planet.

[Our Mission](#)

Making Sustainable Travel Effortless

SmartTrip Planner was born from a simple belief: travel should be both

9 Frontend-Backend Connection Proof

Network tab showing successful API communication

The screenshot below shows the dashboard alongside the browser DevTools Network tab, proving that the React frontend successfully communicates with the FastAPI backend via REST APIs. All requests return HTTP 200 status codes, confirming successful data exchange.

The screenshot displays the SmartTrip AI Planner dashboard and the browser DevTools Network tab. The dashboard shows a welcome message for 'Srishti' and a travel overview for June 2026. It includes four main metrics: 6 Total Trips (+12%), 6 Destinations (+8%), 14.0t CO₂ Saved (+23%), and a 94 Eco Score (+5pt). Below these are two charts: 'Trip Activity' showing monthly trips and eco score trends, and 'Eco Impact' showing a sustainability breakdown with a 94/100 score. The DevTools Network tab shows a list of network requests, all with a status of 200.

Name	Status	Type	Size
login	200	fetch	312 B
me	200	fetch	238 B
stats	200	fetch	415 B
activity	200	fetch	842 B
trips	200	fetch	1.2 kB

✅ **Conclusion:** The React frontend successfully fetches data from the FastAPI backend via authenticated REST API calls. All CRUD operations (Create, Read, Update, Delete) are fully functional with JWT-secured endpoints.